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Constructing Photocatalytic Covalent Organic Frameworks with Aliphatic Linkers

Ting Xu,[⊥] Zhiqiang Wang,[⊥] Weiwei Zhang,* Shuhao An, Lei Wei, Shaomeng Guo, Yanlin Huang, Shan Jiang, Minghui Zhu, Yue-Biao Zhang, and Wei-Hong Zhu*

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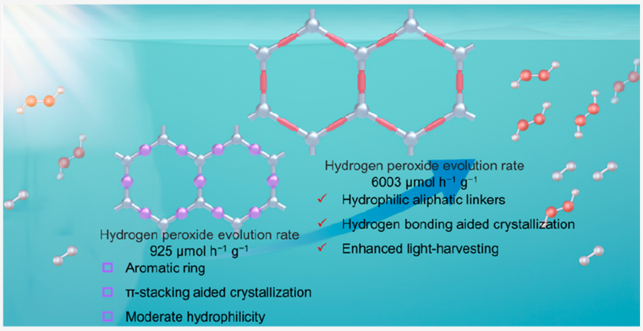
ABSTRACT: Photocatalytic covalent organic frameworks (COFs) are typically constructed with rigid aromatic linkers for crystallinity and extended π-conjugation. However, the essential hydrophobicity of the aromatic backbone can limit their performances in water-based photocatalytic reactions. Here, we for the first time report the synthesis of hydrophilic COFs with aliphatic linkers [tartaric acid dihydrazide (TAH) and butanedioic acid dihydrazide] that can function as efficient photocatalysts for H_2O_2 and H_2 evolution. In these hydrophilic aliphatic linkers, the specific multiple hydrogen bonding networks not only enhance crystallization but also ensure an ideal compatibility of crystallinity, hydrophilicity, and light harvesting. The resulting aliphatic linker COFs adopt an unusual ABC stacking, giving rise to approximately 0.6 nm nanopores with an improved interaction with water guests. Remarkably, both aliphatic linker-based COFs show strong visible light absorption, along with a narrow optical band gap of ~1.9 eV. The H_2O_2 evolution rate for TAH-COF reaches up to 6003 μmol h^{-1} g^{-1}, in the absence of sacrificial agents, surpassing the performance of all previously reported COF-based photocatalysts. Theoretical calculations reveal that the TAH linker can enhance the indirect two-electron oxygen reduction reaction for H_2O_2 production by improving the O_2 adsorption and stabilizing the *OOH intermediate. This study opens a new avenue for constructing semiconducting COFs using nonaromatic linkers.



■ INTRODUCTION

Two-dimensional covalent organic frameworks (2D COFs)[1−3] have emerged as a novel category of porous and crystalline organic materials, becoming a major platform in various applications including gas separation, catalysis, and electronics.[4−12] By precisely integrating tailored organic motifs into layered stacking sequences, the photophysical properties of 2D COFs can be effectively tuned, and this renders them an ideal platform for photoenergy conversion.[13−17] They have demonstrated an immense potential in driving photocatalytic reactions like hydrogen, hydrogen peroxide evolution, and carbon dioxide reduction.[18−27] In general, 2D COFs are constructed by using rigid aromatic linkers. The aromatic stacking favors the periodic spatial organization of building blocks and promotes crystallization of the framework. The formed columnar π arrays could also facilitate exciton migration and charge transport across the molecular skeleton to the catalyst surfaces. Accordingly, attaining full π-conjugation has long been deemed as a fundamental prerequisite for a photocatalytic COF.[28−32] However, most aromatic linkers are relatively hydrophobic, and the resulting framework could suffer from an inadequate affinity for water, limiting their performance in water-based photocatalytic reactions.

As opposed to rigid aromatic linkers, some naturally occurring ligands such as amino acids, carboxylic acids, and their derivatives are more hydrophilic due to the presence of numerous polar functional groups, including hydroxyl, amino, and carbonyl moieties in their side chains.[33−40] These monomers are easily accessed, making them attractive for constructing COFs. However, due to the flexible conformation of the aliphatic chains, incorporating these linkers during synthesis often generates disordered and nonporous structures.[41−44] Another challenge arises from the fact that the incorporation of an aliphatic chain may disrupt the π-conjugation within the molecular framework,[42] leading to an increased optical band gap and the loss of visible light absorption, both undesirable for a photocatalytic COF.

Here, we report the synthesis of two hydrophilic aliphatic linker-based COFs (TAH-COF and BAH-COF) that exhibit exceptionally high photocatalytic activity for H_2O_2 and H_2

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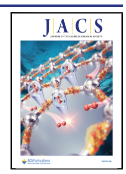

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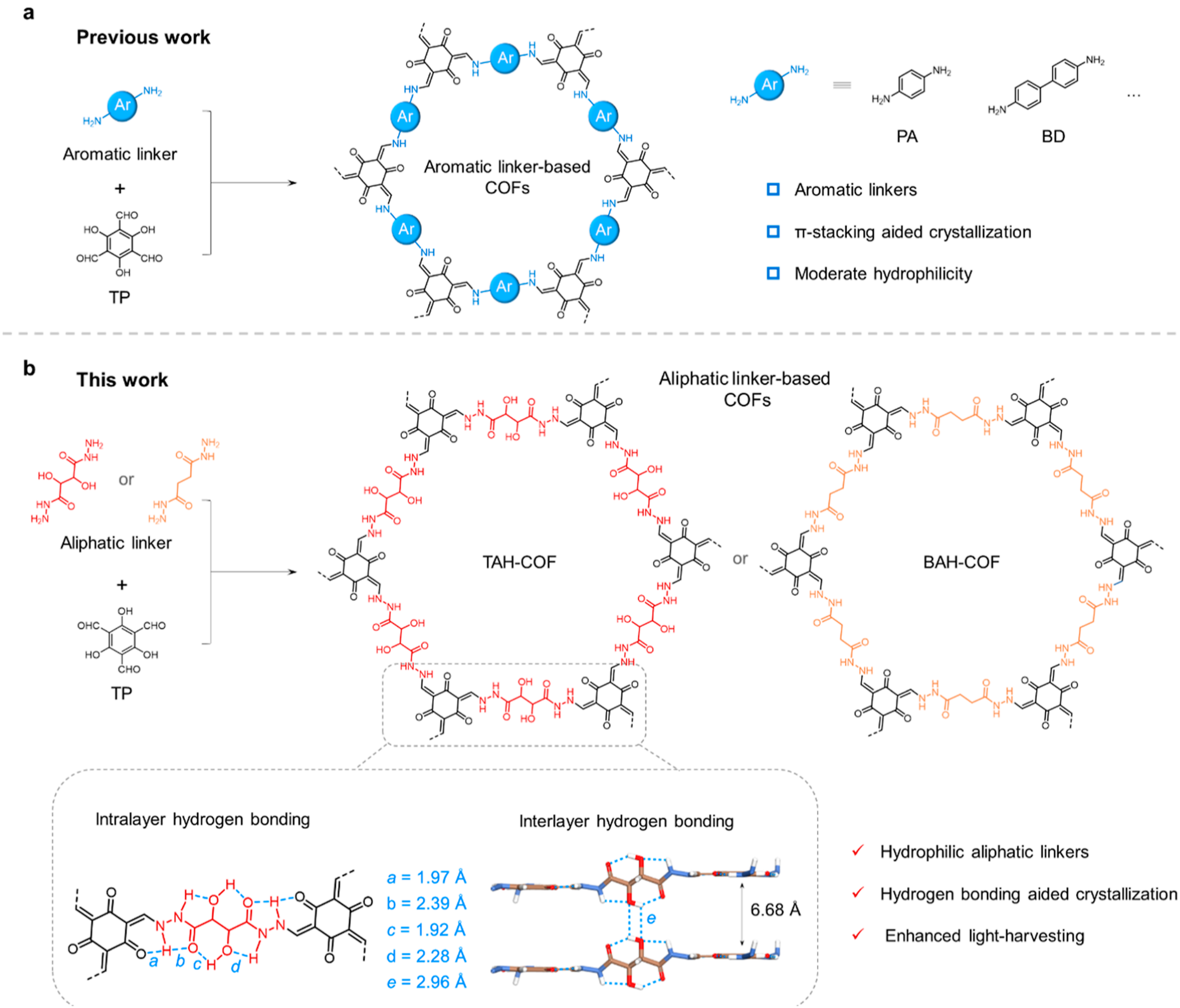

Figure 1. Construction of hydrophilic aliphatic linker COFs. (a) Conventional photocatalytic COFs typically constructed with aromatic linkers. (b) Incorporation of hydrophilic aliphatic linkers (TAH and BAH) into COFs that are crystallized with the assistance of intra- and interlayer hydrogen bonding.

generation. The appropriate proton donors and acceptors on aliphatic linkers can result in specific multiple hydrogen bonding networks, thereby restricting the bond rotation of aliphatic chains and enhancing the framework crystallinity. These aliphatic linker COFs adopt an unusual ABC stacking, giving the formation of approximately 0.6 nm nanopores with improved interaction with water guests. Both COFs show strong visible light absorption properties, and the estimated optical gap of TAH-COF was even narrower than that of the referenced aromatic TPPA-COF with π-conjugated p-phenylenediamine linkers. Remarkably, TAH-COF showed a record-high photocatalytic activity of 6003 μmol h^{-1} g^{-1} for H_2O_2 evolution in the absence of sacrificial agents, together with a solar-to-chemical conversion (SCC) efficiency of 0.66%. Density functional theory (DFT) calculations indicated that the aliphatic TAH linker can enable strong O_2 adsorption and effectively stabilize the intermediates, thereby affording an efficient indirect two-electron ($2e^-$) oxygen reduction reaction (ORR) pathway for H_2O_2 production.

RESULTS AND DISCUSSION

Synthesis of COFs with Hydrophilic Aliphatic Linkers. Novel and functional linkers are always being pursued for tailoring both the chemical properties and the topologies of COFs. Tartaric acid is an important organic compound that exists in nature. Given that the presence of two pair of hydroxyl and carbonyl groups alongside the aliphatic main chain can restrict its conformation variation via intramolecular hydrogen bonding,[36] here we sought to construct COFs using tartaric acid dihydrazide (TAH) and butanedioic acid dihydrazide (BAH) as aliphatic linkers. By condensation with 1,3,5-triformylphloroglucinol (TP) knot, two hydrazone-linked COFs, termed as TAH-COF and BAH-COF (Figure 1b), were generated. The formed imine bond could undergo an irreversible enol-to-keto transformation, generating the β-ketoenamine linkage with an intramolecular hydrogen bonding between keto (C=O) and enamine (N−H) groups.[45] Fourier transform infrared (FT-IR) spectra revealed that the characteristic $\nu_{C=O}$, $\nu_{C=C}$, and ν_{C-N} stretching bands appeared at 1628,

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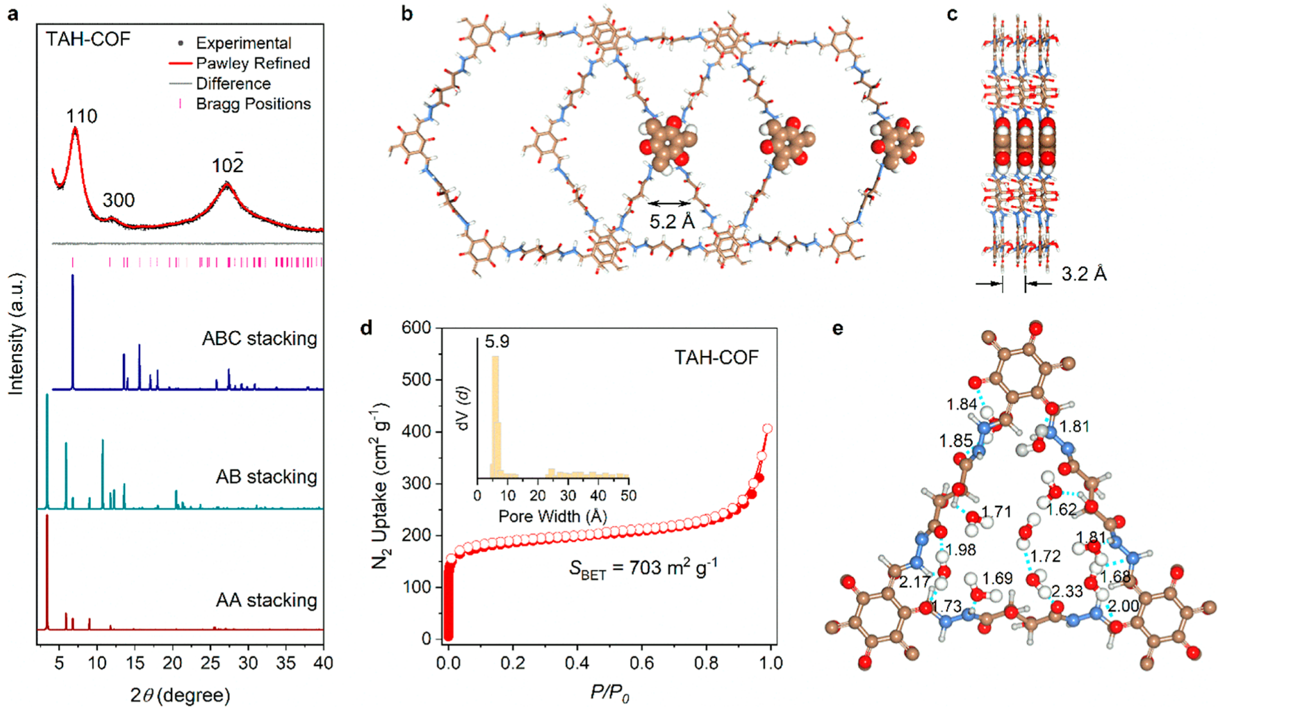

Figure 2. Crystallinity and hydrophilic nanopores. (a) PXRD patterns of TAH-COF: experimental (black), Pawley refined (red), and the corresponding difference (gray); simulated patterns for ABC stacking (blue), AB stacking (green), and AA stacking (brown). Reflection positions are shown by tick marks (pink). Top (b) and side (c) views of the corresponding refined 2D crystal structures of TAH-COF. (d) Nitrogen adsorption−desorption isotherms of TAH-COF measured at 77 K. Inset shows pore size distribution profiles. (e) Top view of hydrophilic pocket in TAH-COF. Color codes: C, brown; N, blue; O, red; and H, white. Hydrogen bonds are depicted as blue dashed lines.

1594, and 1281 cm^{-1} for TAH-COF, respectively, suggesting the formation of a β-ketoenamine linkage (Figures S1−S3). Solid-state ^{13}C cross-polarization magic angle spinning nuclear magnetic resonance (CP-MAS NMR) showed carbonyl and secondary amine carbon atoms at ca. 181 and 150 ppm, respectively, and the resonance signal at ca. 162 ppm was assigned to the carbonyl carbon atom of the alkyl chain (Figure S4). Both COFs possess high physicochemical stabilities and can retain their crystallinities in common solvents for at least 24 h (Figures S5−S8). Scanning electron microscopy and transmission electron microscopy images revealed that TAH-COF exhibited uniform belt-like crystallites with an average size of approximately 600 nm, while BAH-COF showed aggregated spherical particles with the size ranging from 200 to 400 nm in diameter (Figures S9−S12).

The crystallinities of TAH-COF and BAH-COF were measured by powder X-ray diffraction (PXRD), indicative of prominent diffraction peaks at 2θ = ~7.1, 12.1, and 27°. To elucidate the periodical structures of the COFs, three types of possible structure models of eclipsed stacking (AA) and staggered stacking (AB and ABC) models were built, and their geometries were optimized using the Forcite module (Figures 2a and S13). As indicated with Pawley refinement, the experimental PXRD patterns of TAH-COF matched well with the simulated patterns of the staggered stacking (ABC) model in the trigonal $R\bar{3}$ space group, and three characteristic peaks at 2θ = ~7.1, 12.1, and 27° were assigned to the 110, 300, and 10-2 reflection planes, respectively. The lattice parameters for the unit cell were determined as $a = b = 26.06$ Å, $c = 6.57$ Å, $\alpha = \beta = 90°$, and $\gamma = 120°$. DFT calculations also suggested significantly lower crystal stacking energy for the ABC stacking mode than those of AB and AA stacking models (Figure S14). BAH-COF adopts a likewise staggered ABC stacking model as that of TAH-COF (Figure S15).

DFT calculations revealed multiple hydrogen bonding networks in TAH-COF and BAH-COF. For BAH-COF, intra- N−H···O hydrogen bonding was noted where the N−H in the hydrazone linkage acts as proton donors, and the intralayer carbonyl groups of the triglyone node and hydrazone serve as proton acceptors. Along the z-axis, hydrogen bonding for N−H in the hydrazone linkage and carbonyl groups in the adjacent layer was discerned. In TAH-COF, the two hydroxyl groups in the main chain serve as both proton donors and acceptors, forming hydrogen bonds with adjacent carbonyl groups and amide groups. As a whole, there are five types of hydrogen bonding in TAH-COF, including intramolecular N−H···O (1.97, 2.39, and 2.28 Å) and O−H···O (1.92 Å) and intermolecular O−H···O (2.96 Å) (Figures 1b and S17). In specific, there are 50 hydrogen bonds per unit cell of BAH-COF, while the number for TAH-COFs increases to 81 owing to the additional pair of hydroxyl groups on the linker (Figure S18 and Table S1). The hydrogen bonding distances are in the range of 1.76−3.04 Å. This hydrogen bonding network, formed with proper proton donors and acceptors on TAH and BAH, can effectively fix the conformation of aliphatic chains, thereby promoting the crystallization of the COF framework. By contrast, condensation of aliphatic chains without proton donors and acceptors and a TP knot can yield only an amorphous polymer, highlighting the importance of appropriate proton donors and acceptors alongside the ordered assembly of aliphatic chains (Figure S19). In addition, the presence of intralayer hydrogen bonding in TAH-COF and BAH-COF was partially verified through FT-IR spectra (Figure S20).

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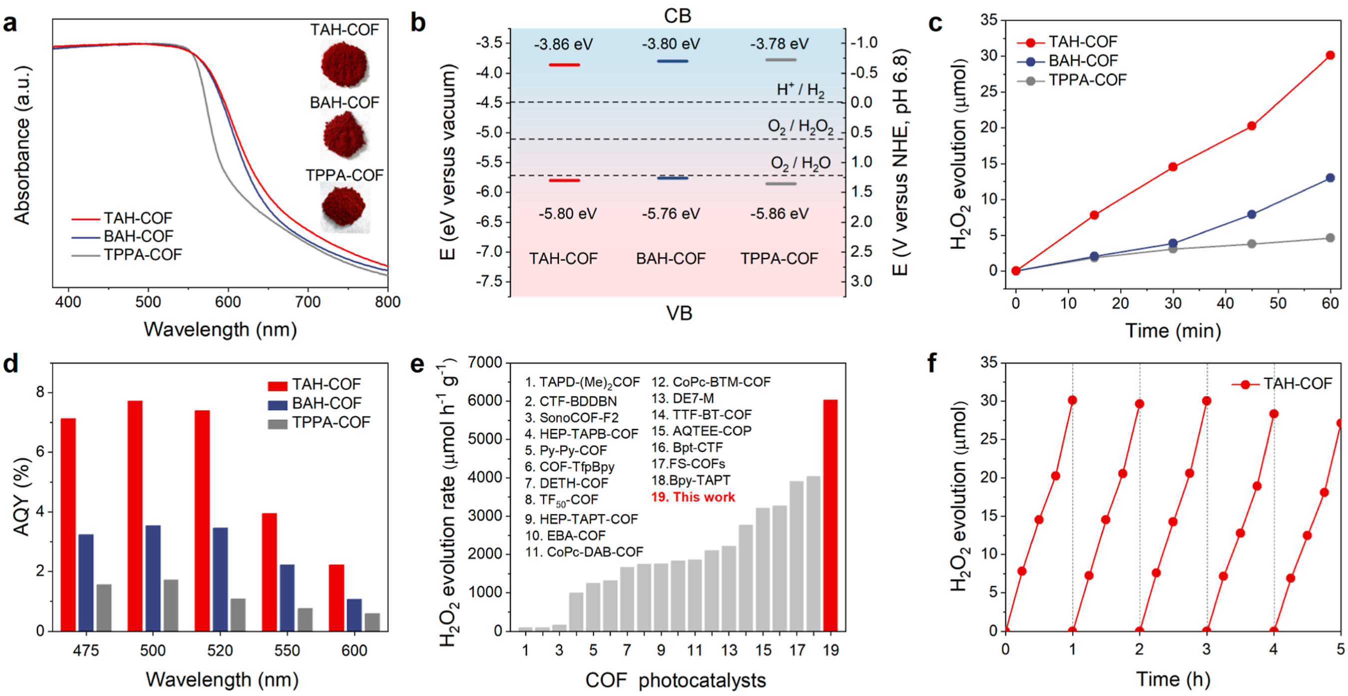

Figure 3. Photophysical properties and photocatalytic H_2O_2 evolution activity. (a) UV−vis absorption spectra for TAH-COF, BAH-COF, and the referenced TPPA-COF measured in the solid state. (b) Band positions of TAH-COF, BAH-COF, and TPPA-COF (these vacuum level values were converted to electrochemical potentials according to −4.5 eV versus vacuum level is equal to 0.0 V versus NHE at pH = 6.8). (c) Time course of H_2O_2 evolution for three COFs (AM 1.5G irradiation, 50 mL of pure water, and 5 mg of COF catalyst). (d) AQYs of TAH-COF, BAH-COF, and TPPA-COF measured at 475, 500, 520, 550, and 600 nm. (e) Summary of photocatalytic H_2O_2 evolution rates of TAH-COF and other reported COF-based photocatalysts; see details in Table S4. (f) Long-term photocatalysis cycling test for TAH-COF.

Hydrophilic and Dynamic Nanopores. The permanent porosities of two aliphatic linker COFs were assessed by measuring N_2 adsorption isotherms at 77 K and CO_2 adsorption isotherms at 273 K. TAH-COF showed a reversible type-I N_2 adsorption isotherm upon activation, and the Brunauer−Emmett−Teller (BET) surface area was evaluated to be 703 $m^2\ g^{-1}$ (Figure 2d). By fitting the nonlocal DFT model to the N_2 adsorption isotherms, the average pore size was centered at 5.9 Å. Similarly, the pore size distributions calculated from CO_2 adsorption isotherms were centered around 5.8 Å, indicating its ultramicroporosity nature (Figure S22). The pore size measured from gas adsorption agreed well with the crystal lattice determined from refined structures, further confirming the formation of staggered ABC stacking. Likewise, BAH-COF has permanent microporosity features with a BET surface area of 530 $m^2\ g^{-1}$, and the regular pore size distribution of around 5.9 Å was consistent with the proposed structural model as well (Figure S23). We noted that the pore sizes of TAH-COF and BAH-COF were the smallest among reported 2D COFs,[46,47] which also makes them attractive candidates for selective molecular/ion adsorption or gas adsorption.

To study the effect of COF linkers on the water affinity of the framework, we further synthesized the state-of-the-art TPPA-COF and TPBD-COF that features an aromatic phenyl linker as a control (Figure 1a). The contact angle measurements with pure water were performed to evaluate the COF/water interface (Figure S24). It was found that the contact angles were markedly decreased from TPBD-COF (65.6°) and TPPA-COF (54.2°) to BAH-COF (47.5°) and TAH-COF (38.9°), indicating an improved water affinity for hydrophilic aliphatic linker COFs. Water vapor adsorption experiments were measured for TAH-COF, BAH-COF, and TPPA-COF at 298 K (Figure S25). The water uptake capacity for TAH-COF and BAH-COF at low partial pressure ($P/P_0 < 0.2$) was found to be significantly higher than that of TPPA-COF, suggesting their stronger water−framework interactions during the pore-filling step. A unique type-IV sorption isotherm was observed for TAH-COF and BAH-COF, which was in contrast to most hydrophobic COFs that typically exhibited a S-shaped type-V sorption curves.[48,49] The computed binding energies also demonstrated stronger interactions for water adsorption in TAH-COF and BAH-COF than in TPPA-COF (Figure S26).

Molecular dynamics simulations were performed to investigate the water adsorption behavior. The staggered stacking model in TAH-COF can provide unique hydrophilic pockets that contain multiple adsorption sites for water (Figure 2e). In instances where a pocket accommodates ten water molecules, nine of them can be involved in hydrogen bonding with the ketone, hydroxyl, and hydrazone sites, and the 10th water molecule forms hydrogen bonds with the previously adsorbed water molecules. The hydrogen bonding distances range from around 1.7 to 2.3 Å, indicating a strong interaction between this pocket and water molecules.[49] Upon achieving adsorption equilibrium, the TAH linker and TP knot undergo noticeable changes in their dihedral angles, decreasing from an initial value of 59.1° to a final value of 0.6° (Figure S28). This adaptive behavior suggested the local flexible nature of framework in the presence of water guests, which can adjust its shape to accommodate the adsorbed water molecules. The hydrophilic nanopore in TAH-COF might be helpful for water-based catalytic reactions.

Intense Visible Light Absorption. We found that both TAH-COF and BAH-COF show intense visible light

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absorption despite using an aliphatic linker and the absorption tails extend beyond 800 nm (Figure 3a). Compared to the referred TPPA-COF constructed with a π-conjugated aromatic linker, TAH-COF and BAH-COF exhibited even red-shifted absorption onsets by 43 and 40 nm, respectively. The optical band gaps were estimated to be 1.94 and 1.96 eV for TAH-COF and BAH-COF, respectively, as compared to 2.08 eV for TPPA-COF (Figures 3b and S29−S32). The enhanced light harvesting for TAH-COF and BAH-COF can be attributed to the resonance-assisted hydrogen bonding (RAHBs),[50,51] where the intramolecular hydrogen bonding facilitates the charge transfer, giving partial delocalization of the π-electrons within the hydrogen bonding motif. This can be further explained by the variation of the band gap for the model compounds and the corresponding COFs (Figure S33). For aromatic linker TPPA-COF, there is a large difference in the optical band gaps for the model compound (MC-TPPA, 2.49 eV) and the formed framework (2.08 eV), indicating that ordered π-stacking and conjugation over the 2D lattice is important for narrowing energy gap and broadening absorption. By contrast, the model compounds MC-TAH and MC-BAH showed very close band gaps to their frameworks, suggesting that the extension of the framework and interlayer stacking have negligible effects on its energy band. Accordingly, the intense visible light absorption was ascribed to the intralayer RAHBs formed between dihydrazide and TP knot.[52] These intralayer RAHBs enhance the polarization in the π-conjugated knot and linkage, narrowing the optical band gap. The slightly red-shifted absorption for MC-TAH compared to that for MC-BAH was due to the presence of additional hydroxy groups that strengthen the RAHB in the main chain.

DFT calculations were employed to determine the highest occupied molecular orbital (HOMO) and lowest unoccupied molecular orbital (LUMO) energy levels for representative fragments of these COFs. The calculated band gap for TAH-COF was 2.16 eV, which is close to that of experimental measurements. The partial density of state analysis for TAH-COF shows that the HOMO is predominantly contributed by C, N, and O atoms, while the LUMO is mostly contributed by the C atoms (Figure S34). The HOMO for TAH-COF was found to be located on the tartaric linkers, while the LUMO was distributed mainly on TP units (Figure S35). The spatially separated HOMO and LUMO in TAH-COF could facilitate electron transfer within the COF frameworks with a lower recombination.

Superior Photocatalytic Activities. The improved hydrophilicity and light-harvesting characteristics of BAH-COF and TAH-COF inspired us to explore its application in photocatalytic reactions. Since their band positions have met the thermodynamics requirements of oxygen reduction to hydrogen peroxide (Figure 3b), we carried out a photocatalytic H_2O_2 production test in pure water, in the absence of sacrificial agents or cocatalysts (Figure 3c). We found that all the three COFs can produce H_2O_2 over 60 min. TPPA-COF showed an average H_2O_2 evolution rate of 925 μmol h^{-1} g^{-1}, while the average evolution rate increased to 1297 μmol h^{-1} g^{-1} for BAH-COF. By comparison, TAH-COF showed a remarkably enhanced H_2O_2 evolution rate up to 6003 μmol h^{-1} g^{-1}, which is 6.5 times higher than that of aromatic TPPA-COF. The presence of two additional hydroxyl groups in the linker of TAH-COF brought forth 4.6 times increase in H_2O_2 evolution rate than that of BAH-COF. The photocatalytic activities for three model compounds are significantly lower than their corresponding COFs, which suggests that the formation of the framework is necessary for COFs to function as good photocatalysts (Figure S37). We extended our investigation by synthesizing benchmark COF-based photocatalysts to compare their photocatalytic H_2O_2 production with that of TAH-COF. Their crystallinities were confirmed through PXRD tests and simulations (Figures S38 and S39). As a result, TAH-COF exhibited remarkable superiority in photocatalytic H_2O_2 evolution (Figure S40). The H_2O_2 production rate for TAH-COF is highest among all reported COF-based photocatalysts thus far (Figure 3e and Table S2).[21−25,31,32,53] This is also the first example of an aliphatic linker COF functioning as a photocatalyst. TAH-COF showed good batch-to-batch reproducibility of photocatalysis (Figure S41).

The apparent quantum yields (AQYs) for TAH-COF were measured at different wavelengths. They were estimated to be 7.12% at 475 nm, 7.72% at 500 nm, 7.39% at 520 nm, 3.94% at 550 nm, and 2.22% at 600 nm (Figure 3d). By contrast, TPPA-COF showed a much lower AQY at the same wavelengths (1.56% at 475 nm, 1.72% at 500 nm, 1.08% at 520 nm, 0.76% at 550 nm, and 0.59% at 600 nm). Overall, TAH-COF achieved a SCC efficiency of 0.66% under AM 1.5G simulated solar irradiation, which is the highest ambient H_2O_2 photosynthesis for COF photocatalysts (Table S2). A long-term H_2O_2 evolution experiment was conducted, and no significant decrease in the catalytic performance was observed upon light irradiation with 5 h (Figure 3f). FT-IR, PXRD, and N_2 adsorption−desorption isotherm measurement indicated the retention of crystalline structures after photocatalysis. The high stability for TAH-COF originated from its chemical rigidity of irreversible β-ketoenamine linkage, together with intra- and interlayer hydrogen bonding networks (Figures S42 and S43).

In light of the high photocatalytic H_2O_2 evolution activity for TAH-COF, we also tested its activity in sacrificial H_2 evolution in water using platinum (Pt) as the cocatalyst. TAH-COF showed an impressive H_2 evolution rate of 31.4 mmol h^{-1} g^{-1}, which is 15.45 times higher than that of TPPA-COF (2.1 mmol h^{-1} g^{-1}) (Figures S44−S46). Indeed, the photocatalytic H_2 evolution activity for TAH-COF was superior to most state-of-the-art COFs reported previously,[13,16,18−20,29,30] noting that these COFs were all constructed with aromatic linkers (Table S3).

The photogenerated charge carrier for TAH-COF was studied by electron spin resonance (ESR) spectroscopy. It exhibited a signal peak at g-factor = 2.003 upon light irradiation, suggestive of an efficient radical generation upon photoexcitation (Figure S47).[54] Light on−off photocurrent experiments showed that TAH-COF responded rapidly to light irradiation and produced an improved photocurrent compared with BAH-COF and TPPA-COF, suggesting more efficient production of charge carriers (Figure S48). Electrochemical impedance spectroscopy results showed that the interfacial charge-transfer resistance of TAH-COF was lower than those of BAH-COF and TPPA-COF (Figure S49). The real-time diffusion and annihilation dynamics of photogenerated free charge carriers was investigated by transient absorption spectra (TAS) (Figure S50). TAH-COF exhibited a significantly longer decay time (18.30 ps) compared to BAH-COF (12.83 ps) and TPPA-COF (0.50 ps), demonstrating slower photogenerated electron−hole recombination.[15] Taken together, the enhanced photocatalysis for TAH-COF can be ascribed to its improved exciton dissociation with long-lived charge carriers and enhanced water affinity.

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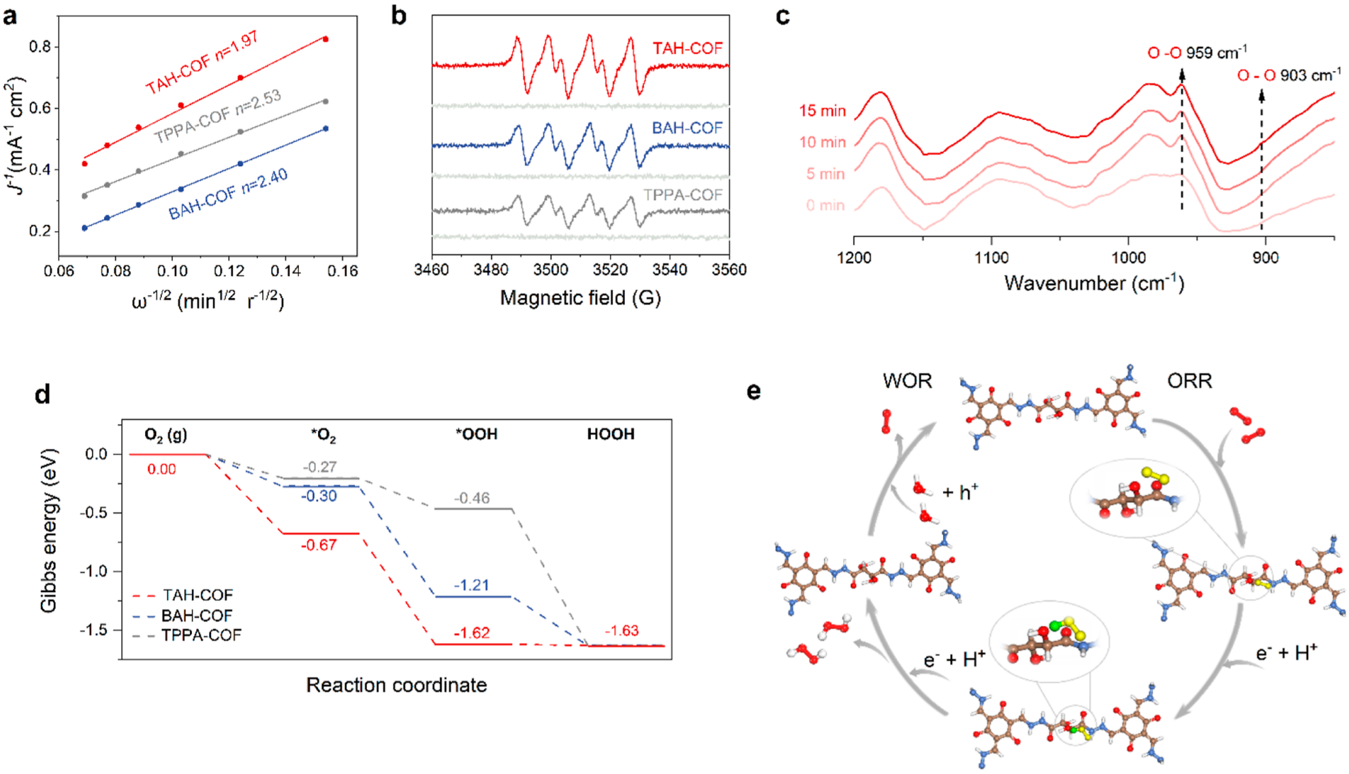

Figure 4. Photocatalytic reaction mechanism. (a) Koutecky−Levich plots for TAH-COF, BAH-COF, and TPPA-COF obtained by RDE tests versus Ag/AgCl. (b) EPR spectra for TAH-COF (red), BAH-COF (blue), and TPPA-COF (black) tested under visible light irradiation (λ > 420 nm) and in the dark (gray) with DMPO as the spin-trap agent. (c) In situ DRIFT spectra as a function of illumination time for TAH-COF under air and water vapor. (d) Free energy change histogram of TAH-COF, BAH-COF, and TPPA-COF for H_2O_2 production. (e) Proposed reaction mechanisms of photocatalytic H_2O_2 production for TAH-COF.

Insight into Photocatalytic H_2O_2 Production. To disclose the photocatalytic reaction pathway of H_2O_2 production, electrochemical measurements were conducted. The rotating disk electrode (RDE) test at different rotating speeds unveiled the average electron-transfer numbers (n) for TAH-COF, BAH-COF, and TPPA-COF were 1.97, 2.40, and 2.53, respectively, demonstrating the 2e$^-$ reduction pathway (Figures 4a and S52). Compared with BAH-COF and TPPA-COF, the number of metastasizing electrons of TAH-COF approaches 2, approving the higher H_2O_2 selectivity under the same conditions. To identify the active oxygen species for the photocatalytic H_2O_2 generation process, electron paramagnetic resonance (EPR) spectroscopy was conducted using 5,5-dimethyl-1-pyrroline N-oxide (DMPO) as a free-radical spin-trapping agent. A typical 6-fold characteristic peak for DMPO$^-$·O^{2-} was observed in the photocatalytic system for these COFs under light irradiation, proving the formation of superoxide radicals (Figure 4b). In situ FT-IR spectrometry was performed to probe the change of the bonding mode in the TAH-COF and the reaction intermediate species during the photocatalytic process (Figure 4c). The intensity of the signals at 903 and 959 cm^{-1} assigned to the O−O bonding gradually increased along with increasing illumination time, confirming the formation of *OOH intermediate species.[23] These characterizations unambiguously prove that TAH-COF reduces the level of aqueous O_2 toward H_2O_2 via a two-step single-electron route. The reaction pathway and corresponding Gibbs free energy differences (ΔG) were further studied by DFT calculations. The adsorption energy for O_2 in TAH-COF is −0.67 eV, which is much lower than that of BAH-COF (−0.27 eV) and TPPA-COF (−0.30 eV) (Figure 4d). The O_2 adsorption isotherms of the three COFs were measured at 273 K, providing further confirmation of the superior O_2 affinity exhibited by TAH-COF (Figure S53). Additionally, the ΔG value of *OOH formation ($^*O_2$ + H^+ + e− → *OOH) for TAH-COF was calculated to be −1.62 eV, which is lower than those of BAH-COF (−1.12 eV) and TPPA-COF (−0.46 eV). These results indicate that O_2 adsorbed on TAH-COF undergoes easier activation to form the *OOH intermediate. Furthermore, it is indicated that the *OOH intermediate can adsorb on the TAH-COF linker by forming two hydrogen bonds with the hydroxyl and amide groups. This stabilization of the *OOH intermediate is beneficial for subsequent protonation to produce H_2O_2 (Figure S54).

To further elucidate the photocatalytic reaction mechanism for TAH-COF, electron and hole scavenger tests were conducted using $KBrO_3$ and benzyl alcohol, respectively (Figure S55). The production of H_2O_2 was inhibited after adding $KBrO_3$, while introducing 10% benzyl alcohol led to a notable increase in overall H_2O_2 production (15,213 μmol h^{-1} g^{-1}). The conversion of alcohol to aldehyde through hole consumption was confirmed by ^{1}H NMR analysis, further demonstrating that the electrons drive this H_2O_2 production. Additionally, to investigate the photooxidation half-reaction, $NaIO_3$ was utilized as an electron scavenger under vacuum conditions. Continuous production of O_2 was observed at a rate of 814 μmol h^{-1} g^{-1}, indicating a 4 h$^+$ water oxidation pathway (Figure S56). Based on these experimental results and DFT calculations, we proposed a photocatalytic H_2O_2 production mechanism for TAH-COF (Figure 4e). Photo-

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excitation of TAH-COF with efficient charge separation leads to the generation of photoexcited e^- and h^+ for the ORR and water oxidation reaction (WOR), respectively. The generated O_2 from the WOR reaction and the dissolved O_2 in water both participated in the ORR process to produce H_2O_2. The efficient charge separation, strong O_2 adsorption ability, and effective stabilization of *OOH intermediates in TAH-COF synergistically promote the indirect $2e^-$ ORR for H_2O_2 production.

■ CONCLUSIONS

In summary, we demonstrate the first example of a photocatalytic COF using aliphatic linkers with unprecedented activity for photosynthetic H_2O_2. The proper proton donors and acceptors on aliphatic linkers guide the formation of hydrogen bonding networks, thus promoting the framework's crystallization. The specific multiple hydrogen bonding networks in the hydrophilic aliphatic linkers can play one stone three birds: (i) assuring hydrophilicity, (ii) guaranteeing crystallinity by avoiding the disordered and nonporous structures, and (iii) enhancing the polarization in the knot and linker to narrow the optical band gap. Specifically, both TAH-COF and BAH-COF adopt an unusual staggered ABC stacking, giving around 0.6 nm polar nanopores with greatly improved water affinity. The RAHBs built in TAH-COF and BAH-COF narrow their band gap and enhance the visible light absorption, endowing these aliphatic linker COFs with semiconducting properties. The ideal combination of crystallinity, hydrophilicity, and light harvesting in TAH-COF brought forth an unprecedented photosynthetic H_2O_2 production rate of 6003 μmol h^{-1} g^{-1} without sacrificial agents, which is the highest among all COF photocatalysts. This study opens a new way to construct semiconducting COFs using aliphatic linkers for solar-to-chemical energy conversions and beyond.

■ ASSOCIATED CONTENT

Supporting Information

The Supporting Information is available free of charge at https://pubs.acs.org/doi/10.1021/jacs.4c04244.

Materials and general methods; experimental details; characterization data including PXRD, FTIR, solid-state ^{13}C NMR, nitrogen, carbon dioxide, and water sorption curves of COFs; electronic properties of COFs; and photocatalytic hydrogen production of COFs (PDF)

Accession Codes

CCDC 2354273 and 2354275−2354277 contain the supplementary crystallographic data for this paper. These data can be obtained free of charge via www.ccdc.cam.ac.uk/data_request/cif, or by emailing data_request@ccdc.cam.ac.uk, or by contacting The Cambridge Crystallographic Data Centre, 12 Union Road, Cambridge CB2 1EZ, UK; fax: +44 1223 336033.

■ AUTHOR INFORMATION

Corresponding Authors

Weiwei Zhang − *Key Laboratory for Advanced Materials and Joint International Research Laboratory of Precision Chemistry and Molecular Engineering, Shanghai Key Laboratory of Functional Materials Chemistry, Feringa Nobel Prize Scientist Joint Research Center, Institute of Fine Chemicals, Frontiers Science Center for Materiobiology and Dynamic Chemistry, School of Chemistry and Molecular Engineering, East China University of Science and Technology, Shanghai 200237, China;* orcid.org/0000-0002-0378-9946; Email: zhangweiwei@ecust.edu.cn

Wei-Hong Zhu − *Key Laboratory for Advanced Materials and Joint International Research Laboratory of Precision Chemistry and Molecular Engineering, Shanghai Key Laboratory of Functional Materials Chemistry, Feringa Nobel Prize Scientist Joint Research Center, Institute of Fine Chemicals, Frontiers Science Center for Materiobiology and Dynamic Chemistry, School of Chemistry and Molecular Engineering, East China University of Science and Technology, Shanghai 200237, China;* orcid.org/0000-0001-9103-166X; Email: whzhu@ecust.edu.cn

Authors

Ting Xu − *Key Laboratory for Advanced Materials and Joint International Research Laboratory of Precision Chemistry and Molecular Engineering, Shanghai Key Laboratory of Functional Materials Chemistry, Feringa Nobel Prize Scientist Joint Research Center, Institute of Fine Chemicals, Frontiers Science Center for Materiobiology and Dynamic Chemistry, School of Chemistry and Molecular Engineering, East China University of Science and Technology, Shanghai 200237, China*

Zhiqiang Wang − *Key Laboratory for Advanced Materials, Centre for Computational Chemistry and Research Institute of Industrial Catalysis, School of Chemistry and Molecular Engineering, East China University of Science and Technology, Shanghai 200237, China;* orcid.org/0000-0001-6293-1474

Shuhao An − *Key Laboratory for Advanced Materials and Joint International Research Laboratory of Precision Chemistry and Molecular Engineering, Shanghai Key Laboratory of Functional Materials Chemistry, Feringa Nobel Prize Scientist Joint Research Center, Institute of Fine Chemicals, Frontiers Science Center for Materiobiology and Dynamic Chemistry, School of Chemistry and Molecular Engineering, East China University of Science and Technology, Shanghai 200237, China;* orcid.org/0000-0001-9099-3314

Lei Wei − *School of Physical Science and Technology, ShanghaiTech University, Shanghai 201210, China*

Shaomeng Guo − *Key Laboratory for Advanced Materials and Joint International Research Laboratory of Precision Chemistry and Molecular Engineering, Shanghai Key Laboratory of Functional Materials Chemistry, Feringa Nobel Prize Scientist Joint Research Center, Institute of Fine Chemicals, Frontiers Science Center for Materiobiology and Dynamic Chemistry, School of Chemistry and Molecular Engineering, East China University of Science and Technology, Shanghai 200237, China*

Yanlin Huang − *School of Physical Science and Technology, ShanghaiTech University, Shanghai 201210, China*

Shan Jiang − *School of Physical Science and Technology, ShanghaiTech University, Shanghai 201210, China*

Minghui Zhu − *State Key Laboratory of Chemical Engineering, School of Chemical Engineering, East China University of Science and Technology, Shanghai 200237, China;* orcid.org/0000-0003-1593-9320

Yue-Biao Zhang − *School of Physical Science and Technology, ShanghaiTech University, Shanghai 201210, China;* orcid.org/0000-0002-8270-1067

Complete contact information is available at:

https://pubs.acs.org/10.1021/jacs.4c04244

Author Contributions

⊥T.X. and Z.W. contributed equally to this work.

Notes

The authors declare no competing financial interest.

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■ REFERENCES

(1) Côté, A. P.; Benin, A. I.; Ockwig, N. W.; O'Keeffe, M.; Matzger, A. J.; Yaghi, O. M. Porous, crystalline, covalent organic frameworks. *Science* **2005**, *310*, 1166−1170.

(2) Evans, A. M.; Strauss, M. J.; Corcos, A. R.; Hirani, Z.; Ji, W.; Hamachi, L. S.; Aguilar-Enriquez, X.; Chavez, A. D.; Smith, B. J.; Dichtel, W. R. Two-dimensional polymers and polymerizations. *Chem. Rev.* **2022**, *122*, 442−564.

(3) Geng, K.; He, T.; Liu, R.; Dalapati, S.; Tan, K. T.; Li, Z.; Tao, S.; Gong, Y.; Jiang, Q.; Jiang, D. Covalent organic frameworks: design, synthesis, and functions. *Chem. Rev.* **2020**, *120*, 8814−8933.

(4) Wang, Z.; Zhang, S.; Chen, Y.; Zhang, Z.; Ma, S. Covalent organic frameworks for separation applications. *Chem. Soc. Rev.* **2020**, *49*, 708−735.

(5) Fenton, J. L.; Burke, D. W.; Qian, D.; Olvera de la Cruz, M.; Dichtel, W. R. Polycrystalline covalent organic framework films act as adsorbents, not membranes. *J. Am. Chem. Soc.* **2021**, *143*, 1466−1473.

(6) Zhao, S.; Jiang, C.; Fan, J.; Hong, S.; Mei, P.; Yao, R.; Liu, Y.; Zhang, S.; Li, H.; Zhang, H.; Sun, C.; Guo, Z.; Shao, P.; Zhu, Y.; Zhang, J.; Guo, L.; Ma, Y.; Zhang, J.; Feng, X.; Wang, F.; Wu, H.; et al. Hydrophilicity gradient in covalent organic frameworks for membrane distillation. *Nat. Mater.* **2021**, *20*, 1551−1558.

(7) Lin, S.; Diercks, C. S.; Zhang, Y.-B.; Kornienko, N.; Nichols, E. M.; Zhao, Y.; Paris, A. R.; Kim, D.; Yang, P.; Yaghi, O. M.; Chang, C. J. Covalent organic frameworks comprising cobalt porphyrins for catalytic CO_2 reduction in water. *Science* **2015**, *349*, 1208−1213.

(8) Liu, S.; Qian, T.; Wang, M.; Ji, H.; Shen, X.; Wang, C.; Yan, C. Proton-filtering covalent organic frameworks with superior nitrogen penetration flux promote ambient ammonia synthesis. *Nat. Catal.* **2021**, *4*, 322−331.

(9) Han, X.; Yuan, C.; Hou, B.; Liu, L.; Li, H.; Liu, Y.; Cui, Y. Chiral covalent organic frameworks: design, synthesis and property. *Chem. Soc. Rev.* **2020**, *49*, 6248−6272.

(10) Wang, M.; Ballabio, M.; Wang, M.; Lin, H.-H.; Biswal, B. P.; Han, X.; Paasch, S.; Brunner, E.; Liu, P.; Chen, M.; Bonn, M.; Heine, T.; Zhou, S.; Cánovas, E.; Dong, R.; Feng, X. Unveiling electronic properties in metal-phthalocyanine-based pyrazine-linked conjugated two-dimensional covalent organic frameworks. *J. Am. Chem. Soc.* **2019**, *141*, 16810−16816.

(11) Qiao, G. Y.; Wang, X.; Li, X.; Li, J.; Geng, K.; Jin, E.; Xu, J.-J.; Yu, J. Unlocking synthesis of polyhedral oligomeric silsesquioxane-based three-dimensional polycubane covalent organic frameworks. *J. Am. Chem. Soc.* **2024**, *146*, 3373−3382.

(12) Keller, N.; Bein, T. Optoelectronic processes in covalent organic frameworks. *Chem. Soc. Rev.* **2021**, *50*, 1813−1845.

(13) Wang, Y.; Vogel, A.; Sachs, M.; Sprick, R. S.; Wilbraham, L.; Moniz, S. J. A.; Godin, R.; Zwijnenburg, M. A.; Durrant, J. R.; Cooper, A. I.; Tang, J. W. Current understanding and challenges of solar-driven hydrogen generation using polymeric photocatalysts. *Nat. Energy* **2019**, *4*, 746−760.

(14) Zhang, W.; Chen, L.; Dai, S.; Zhao, C.; Ma, C.; Wei, L.; Zhu, M.; Chong, S. Y.; Yang, H.; Liu, L.; Bai, Y.; Yu, M.; Xu, Y.; Zhu, X.-W.; Zhu, Q.; An, S.; Sprick, R. S.; Little, M. A.; Wu, X.; Jiang, S.; Wu, Y.; Zhang, Y.-B.; Tian, H.; Zhu, W.-H.; Cooper, A. I. Reconstructed covalent organic frameworks. *Nature* **2022**, *604*, 72−79.

(15) Feng, T.; Streater, D.; Sun, B.; Duisenova, K.; Wang, D.; Liu, Y.; Huang, J.; Zhang, J. Tuning photoexcited charge transfer in imine-linked two-dimensional covalent organic frameworks. *J. Phys. Chem. Lett.* **2022**, *13*, 1398−1405.

(16) Banerjee, T.; Podjaski, F.; Kröger, J.; Biswal, B. P.; Lotsch, B. V. Polymer photocatalysts for solar-to-chemical energy conversion. *Nat. Rev. Mater.* **2021**, *6*, 168−190.

(17) Gong, Y.-N.; Zhong, W.; Li, Y.; Qiu, Y.; Zheng, L.; Jiang, J.; Jiang, H.-L. Regulating photocatalysis by spin-state manipulation of cobalt in covalent organic frameworks. *J. Am. Chem. Soc.* **2020**, *142*, 16723−16731.

(18) Wang, X.; Chen, L.; Chong, S. Y.; Little, M. A.; Wu, Y.; Zhu, W.-H.; Clowes, R.; Yan, Y.; Zwijnenburg, M. A.; Sprick, R. S.; Cooper, A. I. Sulfone-containing covalent organic frameworks for photocatalytic hydrogen evolution from water. *Nat. Chem.* **2018**, *10*, 1180−1189.

(19) Li, C.; Liu, J.; Li, H.; Wu, K.; Wang, J.; Yang, Q. Covalent organic frameworks with high quantum efficiency in sacrificial photocatalytic hydrogen evolution. *Nat. Commun.* **2022**, *13*, 2357.

(20) Ghosh, S.; Nakada, A.; Springer, M. A.; Kawaguchi, T.; Suzuki, K.; Kaji, H.; Baburin, I.; Kuc, A.; Heine, T.; Suzuki, H.; Abe, R.; Seki, S. Identification of prime factors to maximize the photocatalytic hydrogen evolution of covalent organic frameworks. *J. Am. Chem. Soc.* **2020**, *142*, 9752−9762.

(21) Wang, H.; Yang, C.; Chen, F.; Zheng, G.; Han, Q. A Crystalline partially fluorinated triazine covalent organic framework for efficient photosynthesis of hydrogen peroxide. *Angew. Chem., Int. Ed.* **2022**, *61*, No. e2022023.

(22) Zhao, W.; Yan, P.; Li, B.; Bahri, M.; Liu, L.; Zhou, X.; Clowes, R.; Browning, N. D.; Wu, Y.; Ward, J. W.; Cooper, A. I. Accelerated synthesis and discovery of covalent organic framework photocatalysts for hydrogen peroxide production. *J. Am. Chem. Soc.* **2022**, *144*, 9902−9909.

(23) Kou, M.; Wang, Y.; Xu, Y.; Ye, L.; Huang, Y.; Jia, B.; Li, H.; Ren, J.; Deng, Y.; Chen, J.; Zhou, Y.; Lei, K.; Wang, L.; Liu, W.; Huang, H.; Ma, T. Molecularly engineered covalent organic frameworks for hydrogen peroxide photosynthesis. *Angew. Chem., Int. Ed.* **2022**, *61*, No. e2022004.

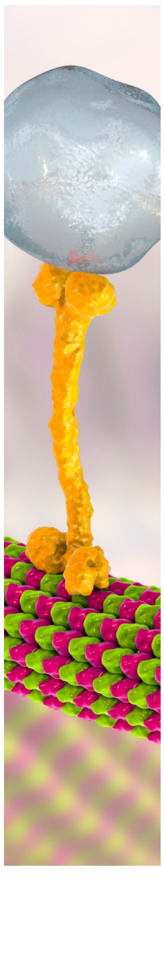
(24) Sun, J.; Sekhar Jena, H.; Krishnaraj, C.; Singh Rawat, K.; Abednatanzi, S.; Chakraborty, J.; Laemont, A.; Liu, W.; Chen, H.; Liu, Y.-Y.; Leus, K.; Vrielinck, H.; et al. Pyrene-based covalent organic frameworks for photocatalytic hydrogen peroxide production. *Angew. Chem., Int. Ed.* **2023**, *62*, No. e202216719.

(25) Krishnaraj, C.; Sekhar Jena, H.; Bourda, L.; Laemont, A.; Pachfule, P.; Roeser, J.; Chandran, C. V.; Borgmans, S.; Rogge, S. M. J.; Leus, K.; Stevens, C. V.; Martens, J. A.; Van Speybroeck, V.; Breynaert, E.; Thomas, A.; Van Der Voort, P. Strongly reducing (diarylamino)benzene-based covalent organic framework for metal-free visible light photocatalytic H_2O_2 generation. *J. Am. Chem. Soc.* **2020**, *142*, 20107−20116.

(26) Lu, M.; Liu, J.; Li, Q.; Zhang, M.; Liu, M.; Wang, J. L.; Yuan, D. Q.; Lan, Y. Q. Rational design of crystalline covalent organic frameworks for efficient CO_2 photoreduction with H_2O. *Angew. Chem., Int. Ed.* **2019**, *58*, 12392−12397.

(27) Zhou, M.; Wang, Z.; Mei, A.; Yang, Z.; Chen, W.; Ou, S.; Wang, S.; Chen, K.; Reiss, P.; Qi, K.; Ma, J.; Liu, Y. Photocatalytic

CO_2 reduction using La-Ni bimetallic sites within a covalent organic framework. *Nat. Commun.* **2023**, *14*, 2473.
(28) Jin, E.; Lan, Z.; Jiang, Q.; Geng, K.; Li, G.; Wang, X.; Jiang, D. 2D sp^2 carbon-conjugated covalent organic frameworks for photocatalytic hydrogen production from water. *Chem* **2019**, *5*, 1632−1647.
(29) Li, Z.; Deng, T.; Ma, S.; Zhang, Z.; Wu, G.; Wang, J.; Li, Q.; Xia, H.; Yang, S.-W.; Liu, X. Three-component donor-π-acceptor covalent-organic frameworks for boosting photocatalytic hydrogen evolution. *J. Am. Chem. Soc.* **2023**, *145*, 8364−8374.
(30) Yang, J.; Ghosh, S.; Roeser, J.; Acharjya, A.; Penschke, C.; Tsutsui, Y.; Rabeah, J.; Wang, T.; Djoko Tameu, S. Y.; Ye, M.-Y.; Grüneberg, J.; Li, S.; Li, C.; Schomäcker, R.; Van De Krol, R.; Seki, S.; Saalfrank, P.; Thomas, A. Constitutional isomerism of the linkages in donor-acceptor covalent organic frameworks and its impact on photocatalysis. *Nat. Commun.* **2022**, *13*, 6317.
(31) Zhi, Q.; Liu, W.; Jiang, R.; Zhan, X.; Jin, Y.; Chen, X.; Yang, X.; Wang, K.; Cao, W.; Qi, D.; Jiang, J. Piperazine-linked metal-phthalocyanine frameworks for highly efficient visible-light-driven H_2O_2 photosynthesis. *J. Am. Chem. Soc.* **2022**, *144*, 21328−21336.
(32) Das, P.; Chakraborty, G.; Roeser, J.; Vogl, S.; Rabeah, J.; Thomas, A. Integrating bifunctionality and chemical stability in covalent organic frameworks via one-pot multicomponent reactions for solar-driven H_2O_2 production. *J. Am. Chem. Soc.* **2023**, *145*, 2975−2984.
(33) Smaldone, R. A.; Forgan, R. S.; Furukawa, H.; Gassensmith, J. J.; Slawin, A. M.; Yaghi, O. M.; Stoddart, J. F. Metal-organic frameworks from edible natural products. *Angew. Chem., Int. Ed.* **2010**, *49*, 8630−8634.
(34) Martí-Gastaldo, C.; Antypov, D.; Warren, J. E.; Briggs, M. E.; Chater, P. A.; Wiper, P. V.; Miller, G. J.; Khimyak, Y. Z.; Darling, G. R.; Berry, N. G.; Rosseinsky, M. J. Side-chain control of porosity closure in single-and multiple-peptide-based porous materials by cooperative folding. *Nat. Chem.* **2014**, *6*, 343−351.
(35) Cheetham, A. K.; Kieslich, G.; Yeung, H. M. Thermodynamic and kinetic effects in the crystallization of metal-organic frameworks. *Acc. Chem. Res.* **2018**, *51*, 659−667.
(36) Alonso, E. R.; León, I.; Kolesniková, L.; Mata, S.; Alonso, J. L. Unveiling five naked structures of tartaric acid. *Angew. Chem., Int. Ed.* **2021**, *60*, 17410−17414.
(37) Sheehan, F.; Sementa, D.; Jain, A.; Kumar, M.; Tayarani-Najjaran, M.; Kroiss, D.; Ulijn, R. V. Peptide-based supramolecular systems chemistry. *Chem. Rev.* **2021**, *121*, 13869−13914.
(38) Dong, J.; Liu, Y.; Cui, Y. Artificial Metal-peptide assemblies: bioinspired assembly of peptides and metals through space and across length scales. *J. Am. Chem. Soc.* **2021**, *143*, 17316−17336.
(39) Bajpayee, N.; Vijayakanth, T.; Rencus-Lazar, S.; Dasgupta, S.; Desai, A. V.; Jain, R.; Gazit, E.; Misra, R. Exploring helical peptides and foldamers for the design of metal helix frameworks: current trends and future perspectives. *Angew. Chem.* **2023**, *135*, No. e202214583.
(40) Jeong, S.; Lee, K.; Yoo, S. H.; Lee, H. S.; Kwon, S. Crystalline metal-peptide networks: structures, applications, and future outlook. *ChemBioChem* **2023**, *24*, No. e202200448.
(41) Koner, K.; Sadhukhan, A.; Karak, S.; Sasmal, H. S.; Ogaeri, Y.; Nishiyama, Y.; Zhao, S.; Polozij, M.; Kuc, A.; Heine, T.; Banerjee, R. Bottom-up synthesis of crystalline covalent organic framework nanosheets, nanotubes, and kippah vesicles: an odd-even effect induction. *J. Am. Chem. Soc.* **2023**, *145*, 14475−14483.
(42) Jiang, S. Y.; Gan, S. X.; Zhang, X.; Li, H.; Qi, Q. Y.; Cui, F. Z.; Lu, J.; Zhao, X. Aminal-linked covalent organic frameworks through condensation of secondary amine with aldehyde. *J. Am. Chem. Soc.* **2019**, *141*, 14981−14986.
(43) Zhang, Y.; Duan, J.; Ma, D.; Li, P.; Li, S.; Li, H.; Zhou, J.; Ma, X.; Feng, X.; Wang, B. Three-dimensional anionic cyclodextrin-based covalent organic frameworks. *Angew. Chem., Int. Ed.* **2017**, *56*, 16313−16317.
(44) Yang, M.; Hanayama, H.; Fang, L.; Addicoat, M. A.; Guo, Y.; Graf, R.; Harano, K.; Kikkawa, J.; Jin, E.; Narita, A.; Müllen, K. Saturated linkers in two-dimensional covalent organic frameworks boost their luminescence. *J. Am. Chem. Soc.* **2023**, *145*, 14417−14426.
(45) Kandambeth, S.; Mallick, A.; Lukose, B.; Mane, M. V.; Heine, T.; Banerjee, R. Construction of crystalline 2D covalent organic frameworks with remarkable chemical (acid/base) stability via a combined reversible and irreversible route. *J. Am. Chem. Soc.* **2012**, *134*, 19524−19527.
(46) Diercks, C. S.; Yaghi, O. M. The atom, the molecule, and the covalent organic framework. *Science* **2017**, *355*, No. eaal1585.
(47) Li, L.; Yun, Q.; Zhu, C.; Sheng, G.; Guo, J.; Chen, B.; Zhao, M.; Zhang, Z.; Lai, Z.; Zhang, X.; Peng, Y.; Zhu, Y.; Zhang, H. Isoreticular series of two-dimensional covalent organic frameworks with the kgd topology and controllable micropores. *J. Am. Chem. Soc.* **2022**, *144*, 6475−6482.
(48) Nguyen, H. L. Covalent organic frameworks for atmospheric water harvesting. *Adv. Mater.* **2023**, *35*, 2300018.
(49) Hanikel, N.; Pei, X.; Chheda, S.; Lyu, H.; Jeong, W.; Sauer, J.; Gagliardi, L.; Yaghi, O. M. Evolution of water structures in metal-organic frameworks for improved atmospheric water harvesting. *Science* **2021**, *374*, 454−459.
(50) Gilli, G.; Bellucci, F.; Ferretti, V.; Bertolasi, V. Evidence for resonance-assisted hydrogen bonding from crystal-structure correlations on the enol form of the .beta.-diketone fragment. *J. Am. Chem. Soc.* **1989**, *111*, 1023−1028.
(51) Liu, B.; Li, J.; Zeng, W.; Yang, W.; Yan, H.; Li, D. C.; Zhou, Y.; Gao, X.; Zhang, Q. High-performance organic semiconducting polymers by a resonance-assisted hydrogen bonding approach. *Chem. Mater.* **2021**, *33*, 580−588.
(52) Qian, H.; Pramanik, S.; Aprahamian, I. Photochromic hydrazone switches with extremely long thermal half-lives. *J. Am. Chem. Soc.* **2017**, *139*, 9140−9143.
(53) Chang, J. N.; Li, Q.; Shi, J. W.; Zhang, M.; Zhang, L.; Li, S.; Chen, Y.; Li, S. L.; Lan, Y. Q. Oxidation-reduction molecular junction covalent organic frameworks for full reaction photosynthesis of H_2O_2. *Angew. Chem., Int. Ed.* **2023**, *62*, No. e202218868.
(54) Jin, E.; Asada, M.; Xu, Q.; Dalapati, S.; Addicoat, M. A.; Brady, M. A.; Xu, H.; Nakamura, T.; Heine, T.; Chen, Q.; Jiang, D. Two-dimensional sp^2 carbon-conjugated covalent organic frameworks. *Science* **2017**, *357*, 673−676.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